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Solving the quartic equation: Ferrari, Euler, and a "modern" approach

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Abstract We discuss the elegant method of Ludovico Ferrari who found a cubic resolvent of the quartic equation in the 16th century. In certain cases, his approach renders the real solutions only in terms of real quantities, which, however, involve up to four nested radicals. A method of Euler brings a reduction to only three nested radicals but makes use of complex quantities. A way to the said cubic resolvent that can be found in many modern textbooks turns out to be less elegant than Ferrari's method.

1 Introduction

The solution of arbitrary cubic equations in one unknown was first published by Gerolamo Cardano in his *Ars magna* of 1545; see [5].

The solution of quartic equations by means of an auxiliary equation of degree 3 (called a *cubic resolvent*) was found by Cardano's student Ludovico Ferrari and also published in the *Ars magna*. More precisely, Ferrari used two different approaches for finding cubic resolvents. The second of these is the simpler one and gives a cubic resolvent of a form that can be found in many modern algebra textbooks. We discuss this elegant method of Ferrari in Section 2.

In Section 3 we study the example used by Cardano in order to explain Ferrari's second approach.

In Section 4 we present a method of Euler that gives the solutions of the quartic equation in terms of only three nested radicals, whereas Ferrari's solution requires four of them. Ferrari's method, however, renders real solutions in terms of (exclusively) real quantities in certain cases, in contrast with Euler's approach.

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In Section 5 we discuss a way to the said cubic resolvent that can be found in many modern textbooks. This method is much more cumbersome than Ferrari's.

2 Ferrari's second approach

We describe only Ferrari's second approach to a cubic resolvent here (see [8, p. 301], [5] (fol. 76)), because its result is simpler than that of the first approach. As concerns this point, the text of [4, p. 509] is misleading, because the discussion of Ferrari's achievements is restricted to the first approach. In [2, pp. 43–45], [11, pp. 397–398], and [9, pp. 321–322] a modification of Ferrari's second approach is given. We briefly highlight Ferrari's first approach and its disadvantage at the end of Section 3.

The starting point is the quartic equation

$$x^4 + ax^2 + c = bx \tag{1}$$

with $b \neq 0$, since, otherwise, the solution of (1) can be reduced to the solution of a quadratic equation. In Cardano's context, a, b, c are positive rational numbers. Equation 1 is general enough, since a possible cubic term can be removed easily; see [2, p. 46].

On adding $a^2/4 - c$ on both sides, we obtain

$$(x^2 + a/2)^2 = bx + a^2/4 - c.$$

Ferrari's idea consists in the introduction of a new unknown t—for which, however, no notation existed at his time; see the end of this section. He wants to have the square of the "trinomial" $x^2 + a/2 + t$, i.e., $(x^2 + a/2 + t)^2$, on the left-hand side. By addition of the respective terms on the right-hand side, he obtains

$$(x^2 + a/2 + t)^2 = bx + a^2/4 - c + 2tx^2 + at + t^2. \tag{2}$$

Under the assumption $t \neq 0$, the right-hand side can be written

$$\left(\sqrt{2t}x + \frac{b}{2\sqrt{2t}}\right)^2 - \frac{b^2}{8t} + \frac{a^2}{4} - c + at + t^2.$$

If we require

$$-\frac{b^2}{8t} + \frac{a^2}{4} - c + at + t^2 = 0, \tag{3}$$

only the square of the expressions in brackets will remain on the right-hand side. Accordingly,

$$\left(x^2 + \frac{a}{2} + t\right)^2 = \left(\sqrt{2t}x + \frac{b}{2\sqrt{2t}}\right)^2. \tag{4}$$

The Eq. 3 can easily be transformed into

$$t^3 + at^2 + (a^2/4 - c)t - b^2/8 = 0. \tag{5}$$

We call (5) *Ferrari's resolvent* of (1). A solution t of the cubic Eq. 5 is different from 0, since $b \neq 0$. Suppose, therefore, that t is such a solution. Then (2) turns into (4), which is equivalent to the equations

$$x^2 + a/2 + t = \pm\left(\sqrt{2t}\,x + \frac{b}{2\sqrt{2t}}\right). \tag{6}$$

In what follows, we assume that (1) has four distinct solutions $x_1, \ldots, x_4$ (which always happens if the polynomial underlying (1) is irreducible). In this case, these solutions satisfy the quadratic Eq. 6. On solving these two equations, we obtain all four solutions of (1), provided that t is a solution of (5). Note that this works for each choice of the solution t.

An appreciation of Ferrari's invention has to take into account that, at his time, it was not yet possible to write the Eq. 1 with *general* coefficients. Indeed, in Cardano's *Ars magna* only equations with *numerical* coefficients appear, for instance in the form "1 qd qd. p: 32 qd. p: 16 aequalia 48 pos.", which stands for "1 bi-square (of the unknown) plus 32 squares (of the unknown) plus 16 are equal to 48 times the unknown"; (see [5] (fol. 76r)). We would write $x^4 + 32x^2 + 16 = 48x$ for this. But letters denoting the unknown do not occur in this book, let alone two letters for two unknowns (in the sense of (2)).

For an online access to the *Ars magna* [5] and to Cossali's book [8] of 1799 see the References. Concerning the early history of the quartic equation, see also [7, pp. 165–178] and [15, pp. 21–24].

3 An example of Cardano

At the end of the foregoing section we mentioned the equation

$$x^4 + 32x^2 + 16 = 48x, \tag{7}$$

which Cardano used to explain Ferrari's method; see [5] (fol. 76r). This equation has two real solutions, say x_1, x_2 and two complex-conjugate solutions, x_3, x_4 (and the polynomial underlying (7) is irreducible).

Cardano says that Eq. 7 arises from the following problem: "Divide 6 into three proportional parts, such that the square of the first plus the square of the second equals 4." This means the following: Denote the three parts by x, y, z. Then

$$x, y, z > 0, \tag{8}$$
$$x + y + z = 6, \tag{9}$$
$$x^2 + y^2 = 4, \tag{10}$$

and

$$x : y = y : z \tag{11}$$

should hold. Problems of this kind were rather common in Renaissance Italy; see [5] (fol. 73v), [16].

From (10) we obtain, since y is positive, $y = \sqrt{4-x^2}$. Then (9) gives $z = 6 - x - \sqrt{4-x^2}$. By (11) , we have $xz = y^2$, and, consequently, $x(6-x-\sqrt{4-x^2}) = 4 - x^2$. Hence (7) easily follows. Ferrari's resolvent (5) in connection with (7) is

$$t^3 + 32t^2 + 240t - 288 = 0; \tag{12}$$

see [5] (fol. 76v). By means of Cardano's formula for the solution of cubic equations (see [17, p. 193]) we obtain the real solution t_1 of (12) in the form

$$t_1 = v_1 + v_2 - 32/3,$$

where

$$v_1 = \frac{2}{3}\sqrt[3]{710 + 18\sqrt{201}}, \quad v_2 = \frac{2}{3}\sqrt[3]{710 - 18\sqrt{201}}, \tag{13}$$

and both third roots are real. The other solutions are $t_2 = v_1\zeta + v_2\zeta^2 - 32/3$ and $t_3 = v_1\zeta^2 + v_2\zeta - 32/3$. Here ζ denotes the primitive 3rd root of unity $(-1+i\sqrt{3})/2$.

If we put $t = t_1$, then the first quadratic Eq. 6 (with the sign "+") yields the real values

$$\sqrt{t_1/2} \pm \sqrt{-t_1/2 - 16 + 24/\sqrt{2t_1}}, \tag{14}$$

which we identify, therefore, with x_1 and x_2. Observe that the real solutions x_1, x_2 are described, exclusively, in terms of real numbers here—which is an advantage (see Section 4). Observe, also, that only the *larger* number of x_1, x_2 yields a solution of (8)–(11).

The reader may ask why $-32/3$ occurs in t_1. This is due to Cardano's formula, which requires the substitution $t = t' - 32/3$. This substitution eliminates the quadratic term in (12). Accordingly, we have to subtract 32/3 from the expression t' given by Cardano's formula.

In the general case, formula (14) reads

$$\sqrt{t_1/2} \pm \sqrt{-t_1/2 - a/2 + b/(2\sqrt{2t_1})}.$$

Here we can eliminate the number t_1 in the denominator. Indeed, by (5),

$$b^2/(8t) = (t + a/2)^2 - c$$

and, consequently, since t_1 is positive,

$$b/(2\sqrt{2t_1}) = \sqrt{(t_1 + a/2)^2 - c}.$$

Probably for this reason the unknown $z = t + a/2$ is introduced instead of t in [2, pp. 43–45].

Cardano uses the example $x^4+6x^2+36 = 60x$ (also considered in [2, pp. 43–45]) in order to display Ferrari's *first* approach (see [5] (fol. 73v)), which we briefly highlight now. We transform the Eq. 1 into

$$(x^2 + \sqrt{c})^2 = (2\sqrt{c} - a)x^2 + bx.$$

Introducing a new unknown u we write

$$(x^2 + \sqrt{c} + u)^2 = (2\sqrt{c} - a + 2u)x^2 + 2\sqrt{c}u + u^2 + bx.$$

We put $A = 2\sqrt{c} - a + 2u$. Then

$$(x^2 + \sqrt{c} + u)^2 = \left(\sqrt{A}x + \frac{b}{2\sqrt{A}}\right)^2$$

holds if $b^2/(4A) = u^2 + 2\sqrt{c}u$. This yields the equation

$$u^3 + (3\sqrt{c} - a/2)u^2 + (2c - a\sqrt{c})u - b^2/8 = 0.$$

In general, this resolvent has two irrational coefficients and is, therefore, less suitable than (5). But in Cardano's example we have $c = 36$ and $\sqrt{c} = 6$. This makes all coefficients rational and the said disadvantage does not catch one's eye. The paper [6] follows Cardano's arguments without mentioning the said disadvantage.

4 Euler's method: elegant but not real

In modern textbooks one frequently finds the following method for solving Eq. 1 (see [18, pp. 135–136], [17, pp. 194–196], [3, pp. 277–278], [13, p. 352]), which is due to Euler; see [10, pp. 309–311], [1, p. 4].

Let t_1 be a solution of Ferrari's resolvent (5). We put $t = t_1$ and consider the solutions of the first Eq. 6 (with the sign "+" on the right-hand side), which we denote by x_1, x_2 (as in the example of Section 3). Then x_3 and x_4 are the solutions of the second Eq. 6. Again, we assume that $x_1, \ldots, x_4$ are distinct. Then the solutions of Ferrari's resolvent (5) are also distinct (which is due to the fact that the discriminants of the left-hand sides of (1) and (5) coincide up to the factor 64; see [18, p. 276]). From (6) we obtain

$$x_1 + x_2 = \sqrt{2t_1}$$

and, thus,

$$(x_1 + x_2)^2 = 2t_1. \tag{15}$$

If we replace t_1 by one of the other solutions of (5), say t_2, the left-hand side of (15) must change. But it still has the form $(x_i + x_j)^2$ for two solutions x_i, x_j of (1), $1 \le i < j \le 4$. Here $\{i, j\} = \{3, 4\}$ is impossible, since (1) implies

$$x_1 + x_2 + x_3 + x_4 = 0 \tag{16}$$

(by Vieta's theorem) and $(x_3 + x_4)^2 = (x_1 + x_2)^2$. Suppose, for instance, $(x_2 + x_4)^2 = 2t_2$. Because $(x_2 + x_4)^2 = (x_1 + x_3)^2$, we obtain $(x_1 + x_3)^2 = 2t_2$. The third solution t_3 of (6) yields the remaining combination $(x_1 + x_4)^2 = 2t_3$. Altogether, we have

$$w_1 = x_1 + x_2 = \sqrt{2t_1}, w_2 = x_1 + x_3 = \sqrt{2t_2}, w_3 = x_1 + x_4 = \sqrt{2t_3}, \tag{17}$$

provided, however, that the roots on the right-hand side have been chosen appropriately (see the end of this section). On using (16), we have

$$w_1 + w_3 + w_3 = 3x_1 + x_2 + x_3 + x_4 = 2x_1 + (x_1 + x_2 + x_3 + x_4) = 2x_1, \tag{18}$$

and

$$w_1 - w_2 - w_3 = x_1 + x_2 - 2x_1 - x_3 - x_4 = 2x_2 - (x_1 + x_2 + x_3 + x_4) = 2x_2. \tag{19}$$

In a similar way

$$-w_1 - w_2 + w_3 = 2x_3 \text{ and } -w_1 + w_2 - w_3 = 2x_4. \tag{20}$$

At Ferrari's time the connections between the solutions and the coefficients of an equation were hardly known, so this way to find the solutions was not yet open.

As concerns the choice of the roots in (17), we find

$$w_1 w_2 w_3 = (x_1 + x_2)(x_1 + x_3)(x_1 + x_4) = x_1^2(x_1 + x_2 + x_3 + x_4) + s_3 = s_3,$$

where s_3 is the elementary symmetric function

$$\sum_{i<j<k} x_i x_j x_k,$$

which equals b; see also the following section.

A choice of the roots w_1, w_2, w_3 in accordance with the relation $w_1 w_2 w_3 = b$ is suitable in the above sense, as can be checked with the help of (16). Here the only issue is the sign of $w_1 w_2 w_3$, which must coincide with the sign of b.

Let us illustrate this with Cardano's example (Section 3). In this example the solution t_1 of (5) ist real and positive, whereas the solutions t_2, t_3 are complex-conjugate. Accordingly, w_1 is real and w_2, w_3 are complex-conjugate. Therefore, the identities (18) and (19) yield the real solutions of (1) and (20) the complex-conjugate ones.

However, even the real solutions x_1, x_2 involve the complex numbers w_2, w_3. This marks a difference between Euler's method and Ferrari's (as discussed in Section 3): Indeed, real solutions are expressed only in terms of real quantities by Ferrari.

So, what is the advantage of Euler's solution, compared with Ferrari's? The answer is as follows: Euler's form of the solution is better with respect to nested radicals. Indeed, consider the solution $t_1 = v_1 + v_2 - 32/3$ of (5). We say, t_1 has *nesting depth* 2 since in formula (13) for v_1 and v_2 we find two nested roots. Accordingly, $\sqrt{2t_1}$ has nesting depth 3 and, in accordance with (14), the solutions x_1 and x_2 have nesting depth 4.

One easily checks that formulas (18)–(20) for the solutions get along with nesting depth 3. However, if we use (18) in order to obtain a form of x_1 that involves only real quantities, we end up with nesting depth 4 again, since the square root of a complex number y involves $|y/|y| + 1|$. The reader may check this.

This situation always occurs—under our assumptions—if (1) has two real and two non-real solutions, which is the same as saying that Ferrari's resolvent (5) has a positive and two non-real solutions; see [18, p. 277].

Since modern textbooks do not discuss Ferrari's approach in most cases, they neither note the reduced nesting depth of Euler's method nor Ferrari's real form of real roots. During the last decades, the problem of reducing the nesting depth of radical expressions aroused considerable interest; for a survey see [14].

5 A "modern" approach to the cubic resolvent

As above, let $x_1, \ldots, x_4$ be the (distinct) solutions of (1). Modern textbooks frequently present a cubic resolvent of the following form:

$$(z-(x_1+x_2)(x_3+x_4))(z-(x_1+x_3)(x_2+x_4))(z-(x_1+x_4)(x_2+x_3)) = 0; \quad (21)$$

see [17, p. 191], [3, pp. 275], [13, p. 352]; slightly modified in [12, p. 261] and [11, pp. 398–399]. If we write

$$g = z^3 + pz^2 + qz + r \quad (22)$$

for the left-hand side of (21), then p, q, r are symmetric polynomials in $x_1, \ldots, x_4$. According to the fundamental theorem of symmetric polynomials (see [17, p. 100], [9, p. 30]) they can be written as polynomial expressions of the elementary symmetric polynomials $s_1 = x_1 + x_2 + x_3 + x_4$, $s_2 = \sum_{i<j} x_i x_j$, $s_3 = \sum_{i<j<k} x_i x_j x_k$, and $s_4 = x_1 x_2 x_3 x_4$. Since the elementary symmetric polynomials agree with the coefficients $0, a, b, c$ of (1) up to the sign, one expresses the coefficients p, q, r

in terms of a, b, c, thereby. This approach can be found in a number of modern textbooks; see [17, p. 193], [3, pp. 276–277], [12, p. 261].

For example, we obtain, by multiplication,

$$-p = (x_1+x_2)(x_3+x_4)+(x_1+x_3)(x_2+x_4)+(x_1+x_4)(x_2+x_3) = 2s_2 = 2a.$$

Expressing q and r in this way is much more complicated. Indeed, multiplication yields

$$q = \sum_{i<j} x_i^2 x_j^2 + 3 \sum_{i\neq j,k;\, j<k} x_i^2 x_j x_k + 6s_4. \tag{23}$$

Further, we obtain, again by multiplication,

$$a^2 = s_2^2 = \sum_{i<j} x_i^2 x_j^2 + 2 \sum_{i\neq j,k;\, j<k} x_i^2 x_j x_k + 6s_4. \tag{24}$$

Due to (16), $s_1 s_3 = 0$ and one finds, once more by multiplication,

$$0 = s_1 s_3 = \sum_{i\neq j,k;\, j<k} x_i^2 x_j x_k + 4s_4. \tag{25}$$

Finally,

$$-4c = -4s_4. \tag{26}$$

On adding up the identities (24), (25), and (26), one obtains

$$a^2 - 4c = \sum_{i<j} x_i^2 x_j^2 + 3 \sum_{i\neq j,k;\, j<k} x_i^2 x_j x_k + 6s_4,$$

which, according to (23), equals q.

We see that the final result must be known in order to establish a reasonable sequence of computational steps. The computation of r is also cumbersome and results in $r = b^2$. Altogether, we have, in the sense of (22),

$$g = z^3 - 2az^2 + (a^2 - 4c)z + b^2. \tag{27}$$

This "modern" resolvent polynomial was already known to François Viète; see [1, p. 3]. The connection between Ferrari's resolvent and g is quite simple. Equation 5 holds, if, and only if, $g=0$ for $z = -2t$.

Summarizing we can say: Ferrari's cubic resolvent (5) ist not substantially different from the equation $g=0$ with g as in (27). However, in modern textbooks the resolvent $g=0$ is established in a way far less elegant than Ferrari's second approach. Probably for this reason some books use this resolvent but do not compute it; see,

for instance, [13, p. 352]. In [11, p. 399] a computer algebra system is recommended for the above computations.

In Weber's *Lehrbuch der Algebra*, which is often considered as a sort of "Bible" of classical algebra, the method for finding this resolvent is more elegant than that of this section, but still laborious; see [18, pp. 135–136]. This method goes back to Euler; see [10, pp. 309–310].

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